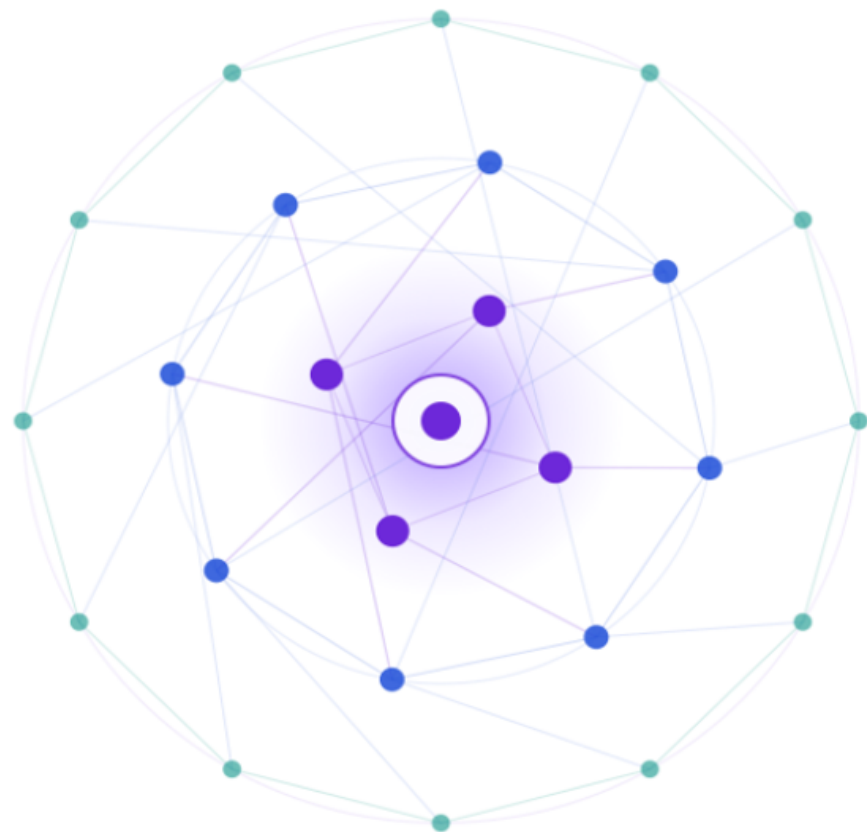


Generative AI Hackathon

3rd Edition 2026 – 27

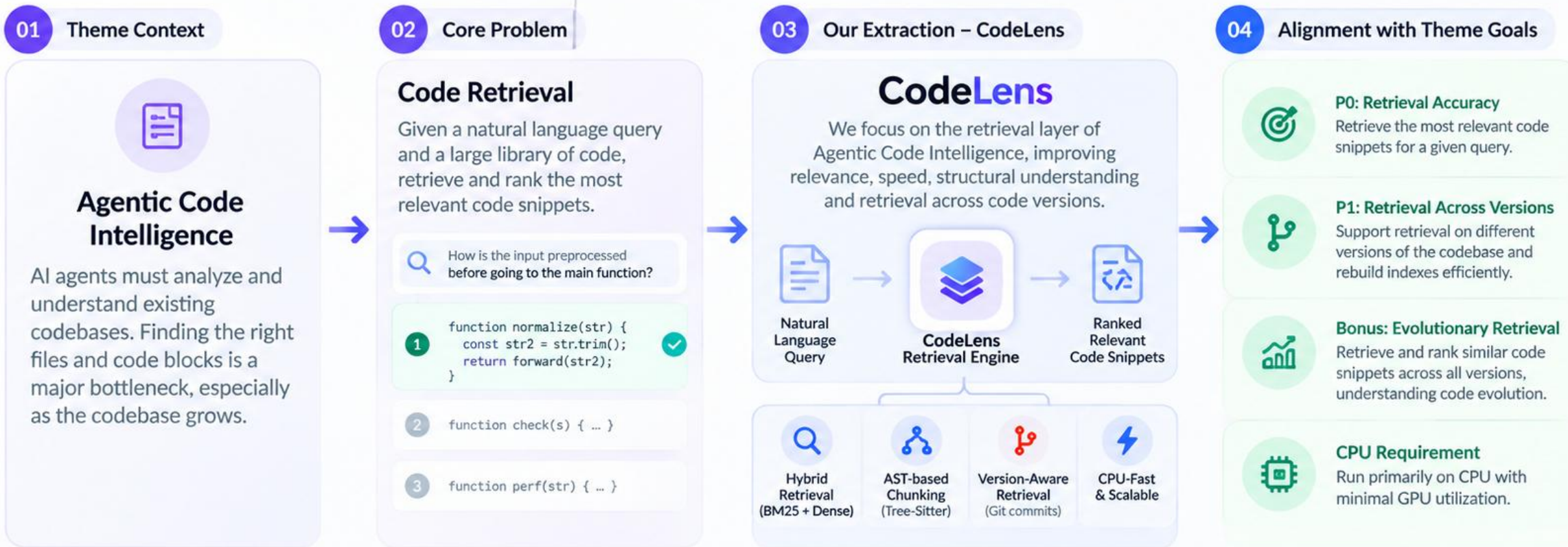
- Theme ID - 1
- Team Name - Currently stuck
- College Name - Ramaiah Institute Of Technology
- Member Name & Email 1- Ashutosh Dwivedi (1ms24ai076@msrit.edu)
- Member Name & Email 2- Ritviz Aggarwal (1ms24ei047@msrit.edu)
- Member Name & Email 3- Aditya Goel (1ms24ec008@msrit.edu)
- Member Name & Email 4- Kriti Sehgal (1ms24cy029@msrit.edu)
- Submission Github link - <https://github.com/AshutoshDwivedi-coder/codelens.git>



Theme

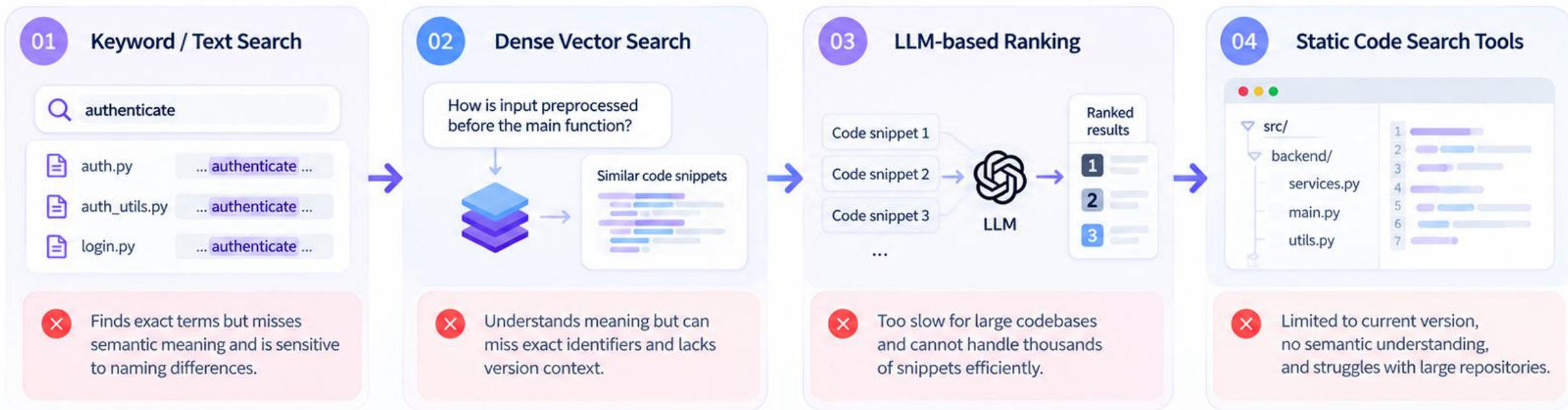
Agentic Code Intelligence

Enable AI agents to understand existing codebases by retrieving the most relevant code snippets for a natural language query, even as the codebase grows and evolves.



Existing Solutions & Gaps

Multiple approaches exist for code search and retrieval, but each has limitations in understanding code semantics, handling large repositories, and retrieving across versions.



Key Gaps

Existing solutions often optimize for one dimension, leaving critical gaps for real-world, large-scale code retrieval.

1 🔍

Retrieval Accuracy (P0)

Often fails to consistently retrieve the most relevant code snippets.

2 🔄

Version Awareness (P1)

Limited support for different versions of the codebase and inefficient re-indexing.

3 ⌚

Evolutionary Retrieval (Bonus)

Struggles to retrieve and rank similar code snippets across versions/commits.

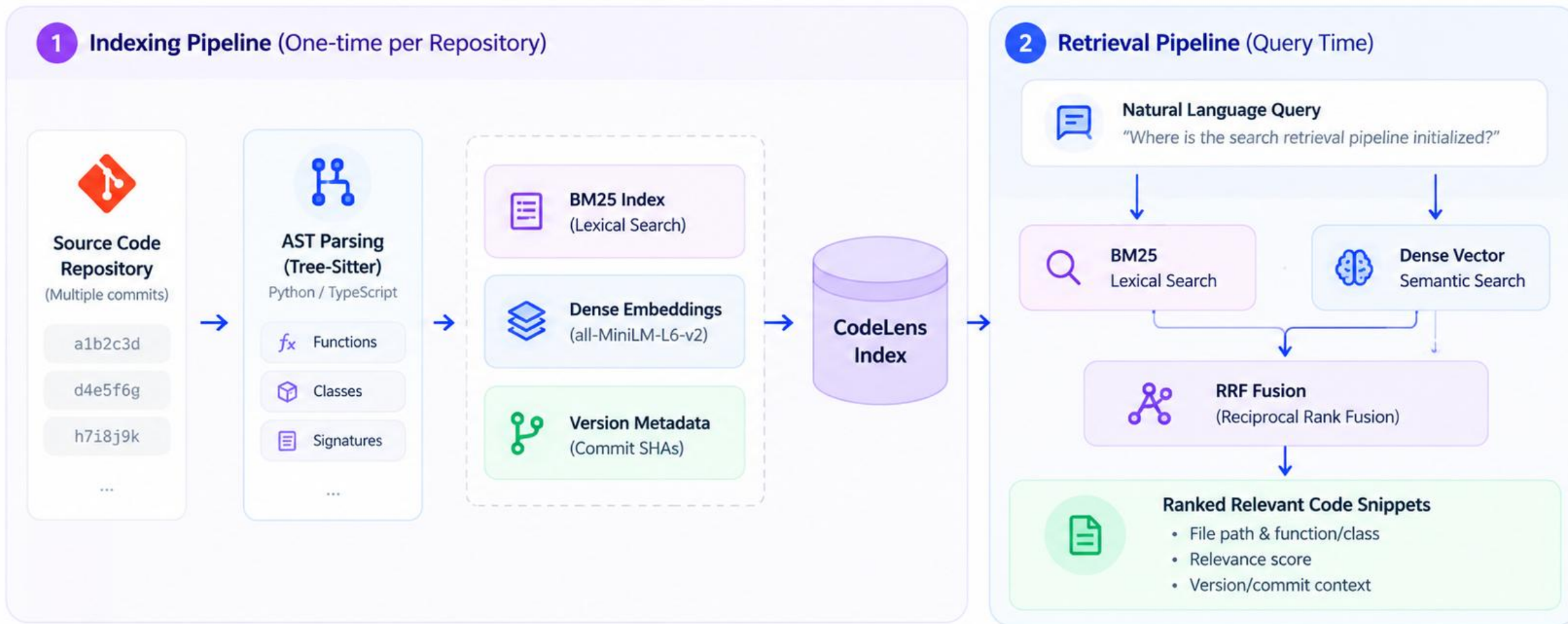
4 🗄️

Scalability & Efficiency

LLM-based methods are slow and cannot handle thousands of snippets; GPU dependency in some approaches.

Our Solutions & Architecture Diagram

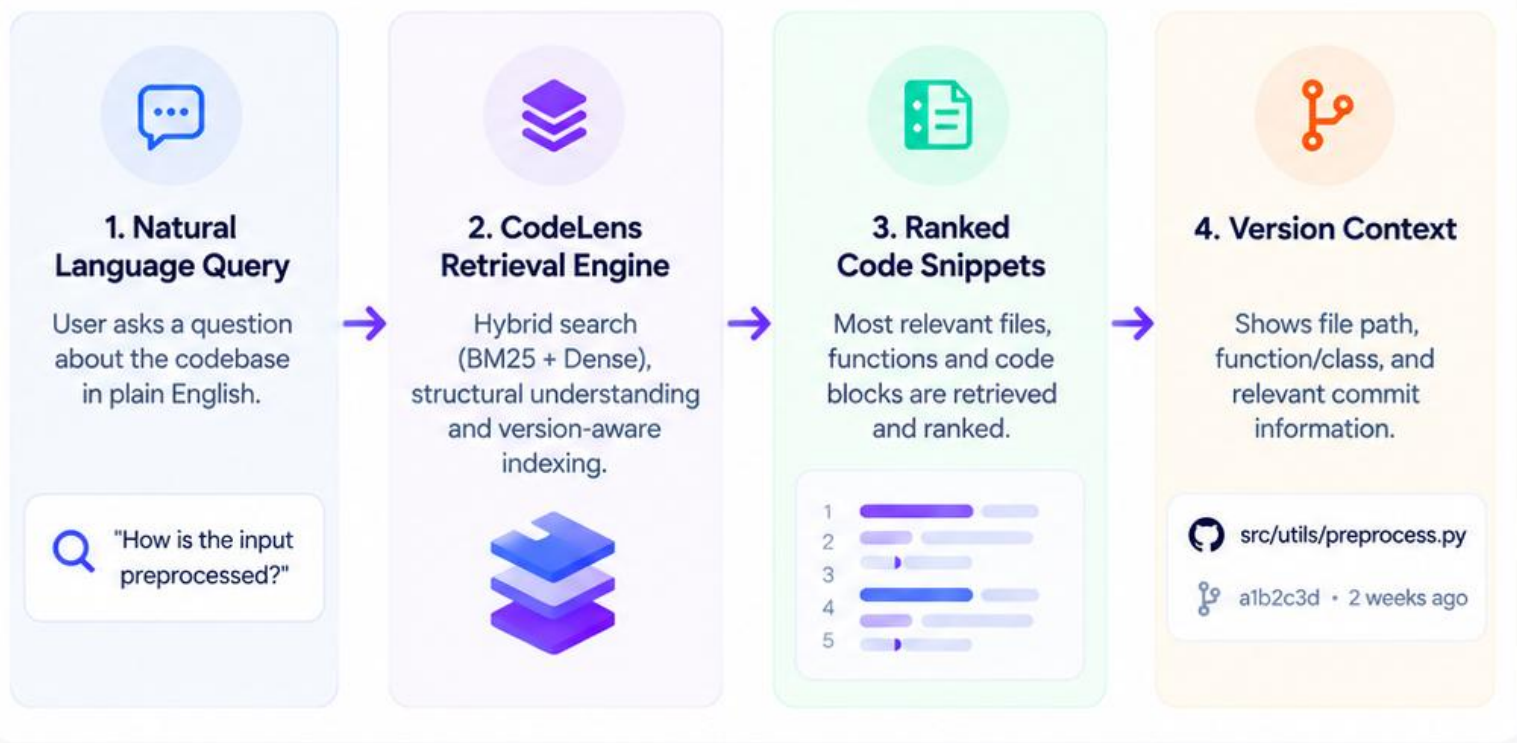
CodeLens uses a hybrid retrieval pipeline that combines structural code understanding, lexical search and semantic embeddings, along with version-aware indexing, to retrieve the most relevant code snippets for a natural language query.



Demo & Product Walkthrough

See how CodeLens retrieves the most relevant code snippets for a natural language query in a real repository, with version-aware and fast retrieval.

1 Product Walkthrough (Flow)



2 Live Demo

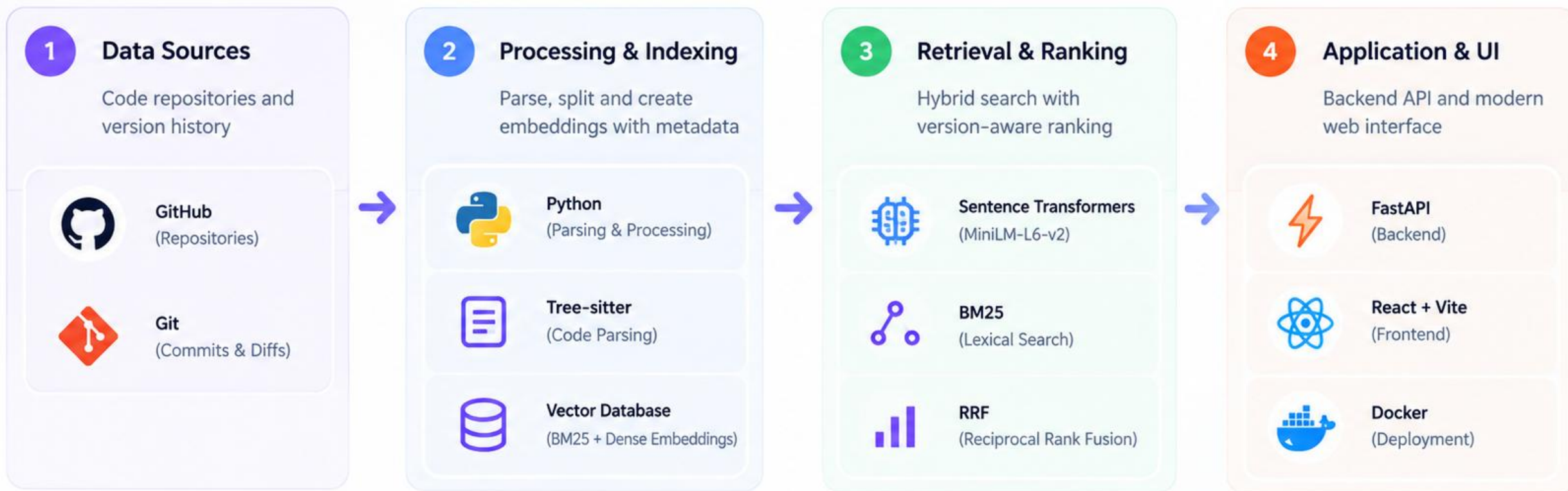
[Live Demo](#)

3 Video Explanation

[Click here](#) 

Tools and tech stack used

A combination of open-source tools and modern technologies to build, index, retrieve and serve code snippets efficiently.



Impact & Use case

Helps developers and teams quickly find, understand, and reuse code, saving time and improving productivity.



Innovation highlights, results and limitations

CodeLens introduces a novel, version-aware hybrid retrieval system that combines code structure, semantics and commit context to deliver more accurate and useful code search.

1 Innovation Highlights



Structure-aware indexing

Understands code structure (AST, functions, classes).



Hybrid retrieval

Combines lexical (BM25) and semantic search for better results.



Version-aware search

Leverages commit history and diffs to find relevant code across versions.



2 Results



Higher relevance

More accurate and context-aware search results.



Faster retrieval

Relevant code snippets in seconds.



Better developer productivity

Spend less time searching, more time building.



3 Limitations & Future Work



Scalability

Performance can degrade for very large repositories.



Language coverage

Currently optimized for popular languages; limited support for niche languages.



Context understanding

May miss complex cross-file or runtime context in some cases.

NEXT STEPS

What's Next?

We will continue to enhance CodeLens with more features, broader support, and better performance to make code search even more powerful and useful.

1 Expand Capabilities



- Support more programming languages
- Handle larger repositories
- Improve understanding of complex code structures

2 Smarter Features



- More advanced semantic search
- Better context awareness
- Personalized and intelligent suggestions

3 Improve Performance



- Faster indexing and retrieval
- Optimized for very large codebases
- Efficient and scalable deployment

4 Broader Adoption



- Integrate with popular dev tools (IDE, Git platforms)
- Enable team collaboration features
- Gather user feedback and keep improving

DIFFERENTIATION

Brownie Points ✨

Extra capabilities that make CodeLens unique and give it an edge over existing tools.

1 Version-aware code search



- Find relevant code snippets across different commits and versions.



2 Hybrid search & intelligent ranking



- Combines semantic, lexical and structural search for more accurate results.



3 Context-aware responses



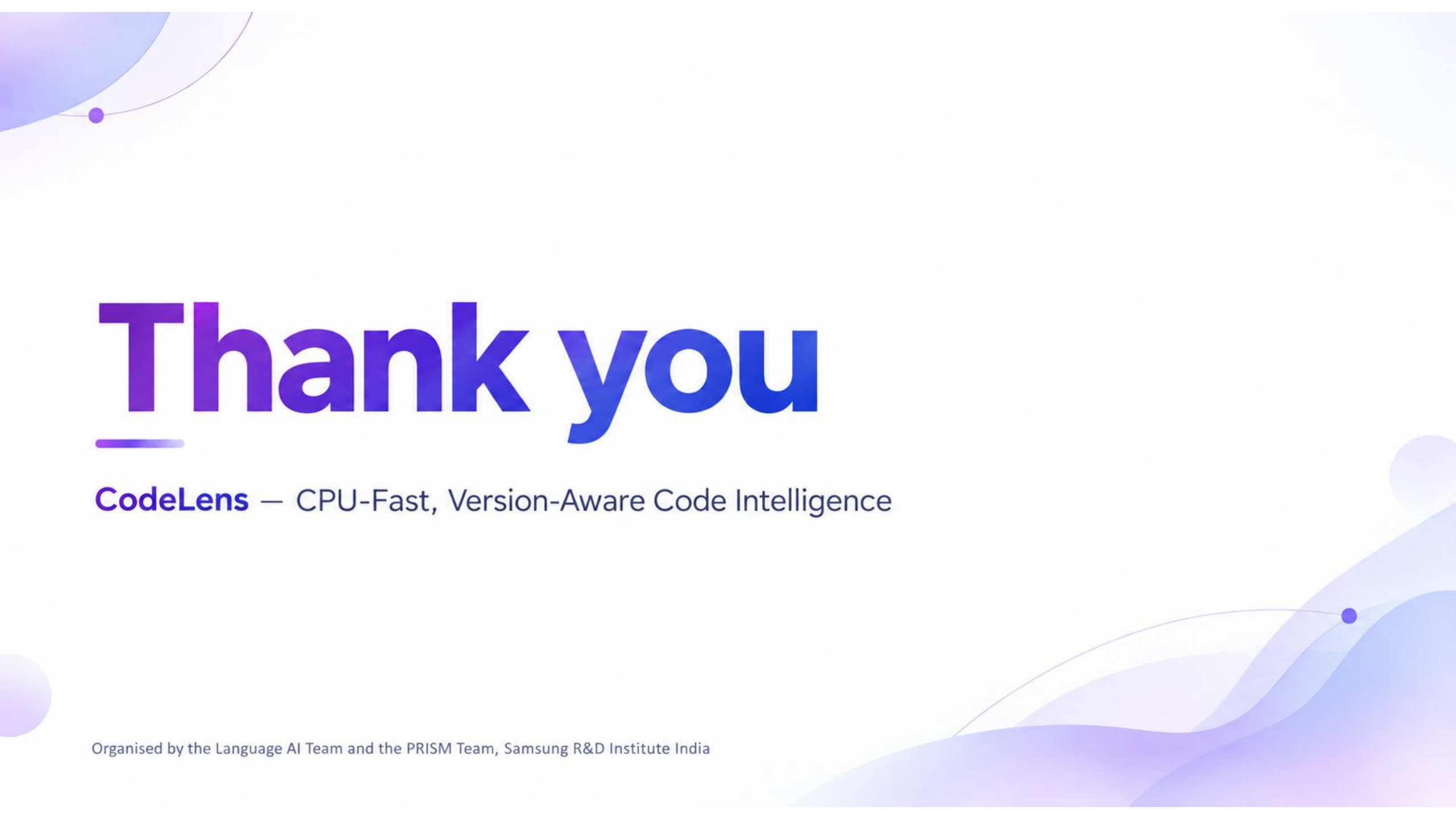
- Shows file path, function context and relevant commit information.



4 Developer-friendly experience



- Clean UI, fast results, and easy integration with popular dev tools.



Thank you

CodeLens — CPU-Fast, Version-Aware Code Intelligence

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India